station at the back of the Humvee. As many as 20 different interfaces were made available to enter data generated by various sensors and systems, like radar from JSTARS (Joint Surveillance Target Attack Radar System) reconnaissance aircraft and video from UAVs.

The system automatically collated all the data on a single electronic map to give commanders a bigger, more accurate and near-real-time picture of the entire battlefield. In its latest avatar, this system uses a Web service architecture to provide near-real information on demand to decision makers in the US, Europe, and the Middle East. This service uses the US military's restricted network called SIPRNET (Secret Internet Protocol Router Network).

A communications system that speaks for the power of "simple" commercial technology is a wearable computer/radio called the TM-7.

It uses an 800 MHz Transmeta Crusoe processor and 256 MB of RAM. Weighing just 5 kg, the device provides mapping, GPS, and communications. With the help of a USB joystick and a helmet-mounted VGA display, soldiers can use TM-7s to control robots—sending them into buildings and caves, avoiding risk to human lives.

Bots And Nanobots

Talking about robots in combat, such a bot—the PackBot—was actually deployed in Iraq and Afghanistan. It entered caves, scouting around and reporting to human operators using wearable computers. In the future, its role could accommodate combat duties: such robots could be loaded with explosives to blow up locations not accessible to soldiers.

Thanks to its 802.11b connection, the PackBot can also be operated over the Internet, allowing for remote operations.

Another DARPA (Defense Advanced Research Projects Agency) funded project, dubbed the High Mobility Tactical Microrobot (HMTM), is in the works. Weighing just 5 pounds, it is being designed for surveillance and reconnaissance. The HMTM has a camera on top of a periscope to look around corners, in addition to an inbuilt homing device that will work even if its 802.11b connection breaks.

We've spoken of Smart Dust earlier in this space; how can it help reduce casualties, which was a primary goal during the Iraq war? The central idea, of course, was to replace people with machines that could gather intelligence. In Iraq, the US military used smart robots and small UAVs to reduce danger to personnel. Another DARPA-supported technology, called Smart Dust, could possibly reduce casualties and gather information even more effectively.

Smart Dust is an "autonomous sensing and communications device in a cubic millimetre" package. A millimetre has not yet been achieved, but the goal is to package a light sensor, power supply and circuitry, a communication device, and a programmable processor into a small space.

Assuming we have it, imagine a plane "spraying" Smart Dust over a conflict area. The specks would be light enough to stay afloat and monitor the movement of enemy troops, or perhaps the presence of biological or chemical weapons. It's not all that futuristic: in a recent test, a Smart Dust researcher controlled a drone about 8 inches long. It flew at 100 kmph for 18 minutes, carrying a camera that sent live feeds back to headquarters!

In Conclusion

Advanced though the above scenarios and technologies may sound, we think they're baby steps into the future of something that will not be called warfare any more. "Warfare" assumes that one is not omniscient; that, like in a game of chess, one needs to outwit the opponent. What happens when sensors and tracking capabilities and number crunching becomes so advanced that everything is *known*? Surrender and victory would both be immediate—removed only a short span of time from the declaration of the conflict. All the activity happens in preparation.

It's analogous to a grandmaster resigning twenty moves into the game once he knows he's made a wrong move—while amateur chess players, like the forces of today, knowing little about the broad picture, take the game into the hundredth move and more. It's about getting smarter—and technology is enabling that. As it is everything else. ■

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